

Rachel Mechum

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EDUCATION

Ph.D. Astrophysics	Rochester Institute of Technology	2023 - ongoing
M.S. Astrophysics	Rochester Institute of Technology	2023 - 2025
B.S. Astrophysics	University of Florida	2021 - 2023

RESEARCH INTERESTS

Gravitational wave data analysis of black hole binary coalescences; Formation of binary black hole systems; Compact binaries in active galactic nucleus disk environments; Origin of supermassive black holes; Galaxy formation and evolution of supermassive black holes using cosmological hydrodynamical simulations

WORK EXPERIENCE

Graduate Research Assistant, Rochester Institute of Technology May 2024 - Present

Utilizing the Rapid Iterative FiTting (RIFT) parameter inference algorithm to investigate waveform systematics in high mass ratio black hole binaries. Additionally developing population synthesis models of binary black hole formation in active galactic nucleus (AGN) disks using the McFACTS framework and BASIL synthetic universe pipeline.

Graduate Teaching Assistant, Rochester Institute of Technology Aug 2023 - May 2024

Aided with two undergraduate courses: Galactic Astrophysics and University Astronomy. Conducted review sessions for exams, assisted with grading assignments such as homework, and offered guidance and support to students navigating academic challenges.

Undergraduate Research Fellow, California Institute of Technology June 2022 - Aug 2022

Selected to participate in the LIGO Summer Undergraduate Research Fellowship program to perform tests of general relativity using the ringdown phase of a binary black hole merger waveform.

Undergraduate Research Assistant, University of Florida Aug 2021 - July 2023

Researched the origin and evolution of supermassive black holes using the IllustrisTNG simulation suite and SMUGGLE model.

PROJECTS

Binary Formation in AGN Disks, Rochester Institute of Technology In Progress

Conducted under the guidance of Dr. Richard O'Shaughnessy, Dr. Saavik Ford, and Dr. Barry McKernan, I am using the McFACTS population synthesis framework to study the formation of high mass ratio binary black holes in the active galactic nucleus (AGN) channel, with a focus on how disk and cluster environments affect their occurrence and detectability.

Master's Thesis, Rochester Institute of Technology [Read Me!](#)

Advised by Dr. Richard O'Shaughnessy, I used the Rapid Iterative FiTting (RIFT) parameter estimation framework to investigate waveform systematics in high mass ratio binary black hole mergers. I analyzed parameter recovery across multiple waveform families, quantifying biases in inferred masses and spins and assessing their impact on gravitational-wave population studies. This work demonstrated the importance of accurate waveform modeling for robust GW inference.

Undergraduate Senior Thesis, University of Florida

[Read Me!](#)

Supervised by Dr. Aklant Bhowmick and Dr. Laura Blecha, I conducted research on the origin and formation of supermassive black holes and how this affects galaxy formation and evolution. Done with the use of the IllustrisTNG simulation suite, Stars and Multiphase Gas in Galaxies (SMUGGLE) model, and the AREPO algorithm. Provided useful information on black hole seed populations and their environments represented by different modelings of the interstellar medium.

LIGO SURF, California Institute of Technology

[Read Me!](#)

Under the mentorship of Dr. Alan Weinstein and graduate student Richard Udall, I investigated tests of general relativity using gravitational wave signals from binary black hole mergers. My work focused on the ringdown portion of the inspiral–merger–ringdown (IMR) waveform, where I extracted higher-order quasinormal modes from simulated merger signals. This analysis demonstrated the importance of including higher order modes when conducting precision tests of general relativity with gravitational wave observations.

PUBLICATIONS

Waveform Systematics of Unequal Mass Binary Black Hole Mergers Using RIFT In prep
R. Mechum, R. O'Shaughnessy.

CONFERENCES, WORKSHOPS, & PRESENTATIONS

APS Global Physics Summit March 2025

Assessing Waveform Systematics for High Mass Ratio Binaries. Poster presentation. Anaheim, CA.

Gravitational Wave Physics Workshop February 2025

Parameter Estimation Using RIFT. Contributed presentation and participant. Syracuse, NY.

AST Jamboree October 2024

A Glimpse into High Mass Ratio Systematics. Formal research presentation. Rochester, NY.

Technical Project Engineering Workshop September 2023

Professional short course in project management. Covered program setup, execution, and closeout, with hands-on program planning. Rochester, NY.

OUTREACH

Research Mentor May 2024 – Present

Mentoring a high school student on an independent project in gravitational wave astrophysics. The project focuses on inspiral–merger–ringdown (IMR) consistency tests, while I provide guidance in analysis methods, communication, and research practices.

Imagine RIT April 2024

Volunteered to organize and run a student-led booth at Imagine RIT, a festival that highlights ongoing and innovative research at the university. Discussed astronomical topics at all educational levels with the greater Rochester community.

Open Observatory Nights 2022-2023

Assisted in organizing and setup of community observing nights with the Astronomy and Astrophysics Society at the University of Florida. Held at the local Rosemary Hill Observatory, I aided with setting up telescopes and making observational astronomy accessible and easily understood to all ages.

AWARDS

Florida Bright Futures Scholarship

2019 - 2023

Selected to receive a scholarship to provide financial aid during undergraduate studies.

SKILLS

Programming & Tools	Python, L ^A T _E X, Git, Bash
Data Analysis & Modeling	RIFT (Rapid Iterative FiTting), McFACTS/BASIL (AGN disk population synthesis), AREPO, IllustrisTNG/SMUGGLE simulations, numerical relativity waveforms, gravitational wave parameter estimation
Research Practices	Statistical inference, simulation analysis, scientific communication, collaboration within the LIGO Scientific Collaboration and cosmological simulation teams
Mentorship & Outreach	Research mentoring of high school students, public science communication, community outreach through observatory events and science festivals